Nuclear and hadronic predictions of the sBootstrap Lagrangian

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Abstract

The sBootstrap $\mathcal{N} = 1$ Lagrangian [3], constructed from SU(3) SQCD at $N_f = N_c = 3$ with an O'Raifeartaigh deformation, makes concrete predictions at the hadronic and nuclear scales. We systematically derive calculational consequences that test the framework against known nuclear physics.

The SUSY pion–muon relation $m_\pi^2 - m_\mu^2 = f_\pi^2$ predicts a SUSY-effective pion decay constant $f_\pi^{\mathrm{SUSY}} = 91.19\,\mathrm{MeV}$, 1.0% below the PDG value. Using this value in the Goldberger–Treiman relation reduces the discrepancy from 1.3% to 0.35%—a factor 3.8 improvement—and predicts $g_{\pi NN} = 13.12$ (0.3% from experiment). The quark condensate derived from the SUSY Gell-Mann–Oakes–Renner relation gives $|\langle\bar{q}q\rangle|^{1/3} = 276\,\mathrm{MeV}$ (6% from the standard lattice value).

The meson spectrum has off-diagonal scalars at $\sqrt{2}\,f_\pi \approx 130\,\mathrm{MeV}$ (pions) and diagonal scalars at $f_\pi \approx 92\,\mathrm{MeV}$. The diagonal mode, lighter than the pion, is identified with the scalar-isoscalar channel and provides a longer-range nuclear attraction (2.2 fm) than pion exchange (1.4 fm), with direct implications for the nuclear force at medium range. Three mesino Dirac pairs at 11, 230, and 490 eV predict a new Yukawa force at 0.4–18 mm range. The muon mass is derived as $m_\mu = \sqrt{m_\pi^2 - f_\pi^2} = 104.9\,\mathrm{MeV}$ (0.7% from experiment), making it a prediction rather than a free parameter. The Seiberg-dual sector predicts $m_d = 4.60\,\mathrm{MeV}$ (0.1σ) and $m_s = 95.0\,\mathrm{MeV}$ (1.9σ).

1 Introduction

A supersymmetric Lagrangian at the nuclear scale was the implicit promise of the programs that connected SUSY with hadronic physics in the 1970s: from Miyazawa's baryon–meson superalgebra [4] through Catto and Gürsey's algebraic SUSY [5] to the modern framework of Seiberg duality [6] and ISS metastable vacua [7]. The sBootstrap [1] makes this promise concrete by providing an explicit $\mathcal{N} = 1$ superpotential and Kähler potential from which the meson, baryon, and mesino spectra follow.

This paper asks: given the Lagrangian of paper [III] [3], what can be *calculated* about nuclear and hadronic physics? Not merely the mass spectrum (already derived in [III]), but the nuclear force, the pion–nucleon coupling, the chiral condensate, the Goldberger–Treiman relation, and other observables that have been the standard tests of any theory of the strong interaction since the 1960s.

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The central tool is the identification $\tilde{m}^2 = f_\pi^2$: the SUSY-breaking soft mass equals the pion decay constant squared. This single identification—fixing the scale of SUSY breaking at $f_\pi \approx 92\,\text{MeV}$—turns the algebraic mass predictions of [III] into dimensional predictions for hadronic observables.

Section 2 derives the SUSY pion decay constant from the pion–muon relation. Section 3 computes the meson spectrum and chiral condensate from the Seiberg vacuum. Section 4 uses the resulting parameters to compute the pion–nucleon coupling, the Goldberger–Treiman relation, and the nuclear potential. Section 5 discusses the muon as a derived quantity and its implications for muonic atoms and muon-catalyzed fusion. Section 6 derives the supertrace sum rule. Section 7 describes the mesino spectrum. Section 8 presents the Seiberg-dual quark mass predictions. Section 9 summarizes the experimental programme.

2 The pion decay constant from SUSY

In the sBootstrap, the pion and the muon belong to the same SUSY multiplet: the pion is a scalar meson M_d^u and the muon is its fermionic partner (a mesino in the SU(2) confining sector). SUSY-breaking splits the masses:

$$m_\pi^2 - m_\mu^2 = f_\pi^2\,. \tag{1}$$

This motivates defining a SUSY-effective pion decay constant:

$$f_\pi^{\text{SUSY}} \equiv \sqrt{m_\pi^2 - m_\mu^2}\,. \tag{2}$$

Evaluating with PDG 2024 [12]:

$$f_\pi^{\text{SUSY}} = \sqrt{(139.57)^2 - (105.66)^2}\,\text{MeV} = \sqrt{8\,316}\,\text{MeV} = 91.19\,\text{MeV}\,, \tag{3}$$

$$f_\pi^{\text{PDG}} = 92.07 \pm 0.57\,\text{MeV}\,, \tag{4}$$

a 1.0% discrepancy. Conversely, assuming the relation is exact, the muon mass is a *derived* quantity:

$$m_\mu^{\text{SUSY}} = \sqrt{m_\pi^2 - f_\pi^2} = \sqrt{19\,480 - 8\,477}\,\text{MeV} = 104.9\,\text{MeV}\,, \tag{5}$$

versus $m_\mu^{\text{exp}} = 105.66\,\text{MeV}$ (0.72% discrepancy). The muon mass is no longer a free parameter: it is determined by m_π and f_π, both QCD observables.

The $\sim 1\%$ precision of (1) admits corrections from: the electromagnetic pion mass (the DGMLY contribution [11] $m_{\pi^\pm}^2 - m_{\pi^0}^2 = 1{,}261\,\text{MeV}^2$), Kähler corrections to the off-diagonal meson mass, and higher-loop CW contributions to the pseudo-modulus.

3 The meson spectrum and chiral condensate

3.1 Meson mass matrix

At the Seiberg vacuum, the meson matrix M acquires a diagonal VEV $\langle M_j^i \rangle = v_i \delta_j^i$ with $v_i \propto 1/m_i$ (the seesaw). The scalar meson fluctuations split into:

Sector	Count	Mass	QCD identification
Off-diagonal M_j^i	12 real	$\sqrt{2}\,f_\pi \approx 130\,\text{MeV}$	$\pi^\pm$, $K^\pm$, K^0, $\bar{K}^0$
Diagonal δM_i^i	6 real	$f_\pi \approx 92\,\text{MeV}$	π^0, η, η', σ

The factor $\sqrt{2}$ between off-diagonal and diagonal masses is the same ratio appearing in the O'Raifeartaigh σ–π splitting at $t = \sqrt{3}$.

The pion mass prediction $m_\pi^{\rm pred} = \sqrt{2}\, f_\pi = 130\,\text{MeV}$ agrees with $m_{\pi^0} = 135\,\text{MeV}$ to 3.7% and with $m_{\pi^\pm} = 139.6\,\text{MeV}$ to 6.9%. The kaon masses (494–498 MeV) are off by a factor of ~ 3.8; this reflects that the leading-order soft mass is flavor-universal while the physical kaon mass is driven by the large $m_s/m_{u,d}$ ratio through the flavor-dependent Kähler potential.

3.2 The diagonal mode as the scalar channel

The six diagonal real modes at $f_\pi \approx 92\,\text{MeV}$ include the scalar-isoscalar combination $(M_u^u + M_d^d)/\sqrt{2}$, which carries the quantum numbers of the σ meson (f_0, $I = 0$, $J^P = 0^+$). At 92 MeV, this mode is lighter than the pion:

$$m_\sigma^{\rm sB} = f_\pi = 92\,\text{MeV}\,, \qquad \text{vs.} \qquad m_\sigma^{\rm QCD} = f_0(500)\text{: pole at 400–550 MeV}\,. \tag{6}$$

The sBootstrap sigma is lighter than the pion and much lighter than the broad $f_0(500)$ resonance. Two interpretations are possible:

1. **Artifact**: the leading-order soft mass does not capture the full dynamics of the scalar-isoscalar channel, which receives large corrections from quark loops, η–η' mixing, and the axial anomaly. In this case, 92 MeV is a tree-level approximation that moves to $\sim 500\,\text{MeV}$ after quantum corrections.

2. **Prediction**: there exists a narrow scalar state near 92 MeV that has escaped detection because its width is suppressed by SUSY relations. This would be a dramatic experimental prediction.

We consider interpretation (1) more likely, since the scalar-isoscalar channel is notorious for large non-perturbative corrections. However, the Yukawa range $\hbar c/m_\sigma^{\rm sB} = 2.16\,\text{fm}$ is the correct order of magnitude for the medium-range nuclear attraction (Section 4).

3.3 The GOR relation and the chiral condensate

The Gell-Mann–Oakes–Renner relation [9]

$$m_\pi^2 f_\pi^2 = -(m_u + m_d)\,\langle \bar{q}q \rangle \tag{7}$$

connects the pion mass to the quark condensate. In the sBootstrap, the off-diagonal meson mass-squared is $2f_\pi^2$ (the soft mass). Substituting:

$$-(m_u + m_d)\,\langle \bar{q}q \rangle = 2 f_\pi^4\,, \tag{8}$$

which gives a parameter-free prediction for the chiral condensate:

$$|\langle \bar{q}q \rangle|^{1/3} = \left(\frac{2 f_\pi^4}{m_u + m_d} \right)^{1/3} = \left(\frac{2 \times (92.07)^4}{6.83} \right)^{1/3} \text{MeV} = 276\,\text{MeV}\,. \tag{9}$$

The standard lattice value is $|\langle \bar{q}q \rangle|^{1/3} \approx 260\,\text{MeV}$ (at $\mu = 2\,\text{GeV}$, $\overline{\text{MS}}$), a 6.2% discrepancy. This is consistent given the leading-order approximation: the $\sqrt{2}$ factor in $m_\pi^{\rm sB} = \sqrt{2} f_\pi$ versus the physical $m_\pi = 135\,\text{MeV}$ propagates into the condensate estimate. With the physical pion mass, GOR gives $|\langle \bar{q}q \rangle|^{1/3} = (m_\pi^2 f_\pi^2/(m_u + m_d))^{1/3} = 261\,\text{MeV}$, recovering the standard value.

4 The nuclear force: GT, $g_{\pi NN}$, and the NN potential

4.1 The Goldberger–Treiman relation

The Goldberger–Treiman (GT) relation [10]

$$g_{\pi NN}\, f_\pi = g_A\, m_N \tag{10}$$

connects the pion–nucleon coupling $g_{\pi NN}$ to the axial coupling $g_A = 1.2756 \pm 0.0013$ (neutron β decay) and the nucleon mass $m_N = 938.3\,\mathrm{MeV}$. With $g_{\pi NN} = 13.17 \pm 0.06$ (Nijmegen partial-wave analysis), the GT discrepancy is

$$\Delta_{\mathrm{GT}} \equiv 1 - \frac{g_A m_N}{g_{\pi NN} f_\pi}\,. \tag{11}$$

Using the PDG pion decay constant $f_\pi = 92.07\,\mathrm{MeV}$:

$$\Delta_{\mathrm{GT}}^{\mathrm{PDG}} = 1 - \frac{1.2756 \times 938.3}{13.17 \times 92.07} = 1 - 0.987 = 1.3\%\,. \tag{12}$$

Using the SUSY-predicted value $f_\pi^{\mathrm{SUSY}} = 91.19\,\mathrm{MeV}$:

$$\boxed{\Delta_{\mathrm{GT}}^{\mathrm{SUSY}} = 1 - \frac{1.2756 \times 938.3}{13.17 \times 91.19} = 1 - 0.997 = 0.35\%\,.} \tag{13}$$

The SUSY-effective pion decay constant reduces the GT discrepancy by a factor of 3.8. This is the single most concrete nuclear prediction of the sBootstrap: the "correct" f_π for the GT relation is $\sqrt{m_\pi^2 - m_\mu^2}$, not the PDG value from $\pi \to \mu\nu$.

4.2 The pion–nucleon coupling

Alternatively, using g_A and m_N as inputs and the GT relation as an *exact* consequence of SUSY, we predict $g_{\pi NN}$:

$$g_{\pi NN}^{\mathrm{PDG}} = \frac{g_A m_N}{f_\pi^{\mathrm{PDG}}} = \frac{1.2756 \times 938.3}{92.07} = 13.00\,, \tag{14}$$

$$g_{\pi NN}^{\mathrm{SUSY}} = \frac{g_A m_N}{f_\pi^{\mathrm{SUSY}}} = \frac{1.2756 \times 938.3}{91.19} = 13.12\,. \tag{15}$$

The experimental value $g_{\pi NN} = 13.17 \pm 0.06$ is reproduced to 0.38% by f_π^{SUSY}, versus 1.3% with f_π^{PDG}.

4.3 The one-pion-exchange nuclear potential

The backbone of nuclear physics is the one-pion-exchange potential (OPEP). For the NN central channel at distance r:

$$V_{\mathrm{OPEP}}^{(\mathrm{central})}(r) = -\frac{g_{\pi NN}^2}{4\pi}\,\frac{m_\pi^3}{12\, m_N^2}\,\frac{e^{-m_\pi r/(\hbar c)}}{m_\pi r/(\hbar c)}\,. \tag{16}$$

At $r = 1\,\mathrm{fm}$, with $g_{\pi NN} = 13.12$ (SUSY) and $m_\pi = 139.6\,\mathrm{MeV}$:

$$V_{\mathrm{OPEP}}(1\,\mathrm{fm}) = -2.5\,\mathrm{MeV}\,. \tag{17}$$

The pion exchange range is $\hbar c/m_\pi = 1.41\,\mathrm{fm}$.

4.4 Medium-range attraction: the diagonal meson exchange

The sBootstrap predicts a scalar-isoscalar mode at $m_\sigma^{\rm sB} = f_\pi = 92\,\text{MeV}$. If this mode couples to nucleons with strength $g_{\sigma NN}$, it contributes a central attractive potential

$$V_\sigma(r) = -\frac{g_{\sigma NN}^2}{4\pi}\,\frac{e^{-m_\sigma r/(\hbar c)}}{r}\,, \tag{18}$$

with range $\hbar c/m_\sigma^{\rm sB} = 2.16\,\text{fm}$—longer than the pion range by a factor 1.53.

In the standard nuclear force paradigm, the medium-range attraction (the "sigma channel") at $r \approx 1$–$2\,\text{fm}$ is provided by two-pion exchange or an effective σ meson at 400–550 MeV (range 0.35–0.49 fm). The sBootstrap places this attraction at a larger range. The Yukawa factor at $r = 1\,\text{fm}$ is $e^{-x}/x = 1.36$ for $m_\sigma = 92\,\text{MeV}$, versus 0.045 for $m_\sigma = 450\,\text{MeV}$: a factor 30× stronger. Even a small coupling $g_{\sigma NN} \sim 1$ would produce a significant medium-range potential.

Whether this longer-range scalar exchange improves or worsens the nuclear force fit is an open question. It may be that the leading-order 92 MeV mode receives large corrections (as argued in Section 3) and moves to $\sim 500\,\text{MeV}$ in the full theory. Alternatively, the extra attraction at 2 fm could account for the intermediate-range NN attraction that in the Bonn/CD-Bonn potentials is parameterized by correlated two-pion exchange with an effective mass of $\sim$ 300–500 MeV; the sBootstrap diagonal mode provides a candidate for the physical origin of this term at a lower mass.

4.5 The deuteron binding: qualitative estimate

The deuteron binding energy $E_d = 2.224\,\text{MeV}$ corresponds to a characteristic size $1/\kappa = 6.1\,\text{fm}$ ($\kappa = \sqrt{m_N E_d/2}/\hbar c = 0.164\,\text{fm}^{-1}$). This is 4.3 times the pion range and 2.8 times the sBootstrap sigma range: the deuteron "lives" in the tail of the nuclear potential, dominated by the lightest exchanged meson.

If the lightest scalar exchange has mass $m_\sigma^{\rm sB} = 92\,\text{MeV}$, the Yukawa tail at the deuteron radius is $e^{-0.92\cdot 6.1/2.16} = e^{-2.6} = 0.074$—not negligible. At the pion mass, $e^{-6.1/1.41} = e^{-4.3} = 0.013$—exponentially suppressed. This suggests that the deuteron binding in the sBootstrap is sensitive to the scalar-isoscalar channel at 92 MeV, and getting E_d right constrains $g_{\sigma NN}$.

A full calculation of E_d from the sBootstrap requires solving the Schrödinger equation with the tensor and central OPEP plus the scalar exchange, and lies beyond the scope of this paper. However, the qualitative point is that the sBootstrap provides *both* the pion and sigma parameters from a single Lagrangian—precisely the information needed to compute nuclear binding.

5 The muon mass and muon-catalyzed fusion

5.1 The muon as a derived quantity

In the sBootstrap, the muon mass is not free: from (5), $m_\mu = \sqrt{m_\pi^2 - f_\pi^2} = 104.9\,\text{MeV}$, a 0.72% prediction. This means the mass hierarchy that makes muon-catalyzed fusion (MCF) possible is a *consequence* of the SUSY-breaking structure:

$$\frac{m_\mu}{m_\pi} = \sqrt{1 - \frac{f_\pi^2}{m_\pi^2}} = \sqrt{1 - \frac{1}{2}\frac{m_\pi^2}{m_\pi^2}} = \sqrt{1 - 0.434} = 0.752\,, \tag{19}$$

using $f_\pi^2/m_\pi^2 \approx 1/2$ (the SUSY relation). The muon is lighter than the pion by the factor $\sqrt{1 - f_\pi^2/m_\pi^2}$, which is fixed by SUSY breaking. A heavier muon (say $m_\mu > m_\pi$) would make MCF impossible; the sBootstrap predicts $m_\mu < m_\pi$ as a structural necessity.

5.2 Muonic hydrogen and the nuclear overlap

The muonic hydrogen Bohr radius

$$a_\mu = \frac{\hbar c}{\alpha\, m_\mu} = 255.9\,\text{fm} \tag{20}$$

is 207 times smaller than the electronic Bohr radius, enabling nuclear overlap. With the SUSY-predicted $m_\mu = 104.9\,\text{MeV}$: $a_\mu^{\text{SUSY}} = 257.8\,\text{fm}$ (shift $+1.9\,\text{fm}$, 0.7%). The 1s binding energy shifts from 2.813 keV to 2.793 keV, a 20 eV change.

For MCF, the key step is the resonant formation of the $dt\mu$ molecular ion, with binding energy $\sim 0.66\,\text{eV}$ in the $J = 1$ state. The 20 eV shift from the SUSY mass prediction is $30\times$ larger than the molecular binding: the SUSY prediction is not precise enough to compute MCF rates from first principles.

However, the SUSY framework provides a different kind of leverage. The GT relation with f_π^{SUSY} gives a more accurate $g_{\pi NN}$, which enters the nuclear potential that governs the d–t interaction inside the molecular ion. The nuclear potential at the internuclear separation of $dt\mu$ ($\sim 1\,\text{fm}$) is dominated by OPEP; a 0.3% improvement in $g_{\pi NN}$ propagates into a corresponding improvement in the nuclear overlap integral.

More generally, the sBootstrap explains *why* MCF works: the muon is lighter than the pion by SUSY, heavy enough to bring nuclei within range of the nuclear force (which is itself set by the pion mass), and the GT relation ties all three scales (m_μ, m_π, f_π, $g_{\pi NN}$, m_N) into a single self-consistent framework.

5.3 The pion–muon mass difference and nuclear stability

The mass difference $m_\pi - m_\mu = 33.9\,\text{MeV}$ is the available energy for pion decay $\pi^+ \to \mu^+\nu_\mu$; its largeness (relative to m_e) determines the pion lifetime and hence the range of the nuclear force in matter. In the sBootstrap:

$$m_\pi - m_\mu = m_\pi \left(1 - \sqrt{1 - \frac{f_\pi^2}{m_\pi^2}}\right) \approx \frac{f_\pi^2}{2\,m_\pi} = \frac{(92.1)^2}{2 \times 139.6}\,\text{MeV} = 30.4\,\text{MeV}\,, \tag{21}$$

versus the observed 33.9 MeV (10% discrepancy, from the 1% error in $f_\pi^{\text{SUSY}}/f_\pi^{\text{PDG}}$ amplified by the nonlinear $\sqrt{1-x}$ function). The pion decay phase space $\propto (m_\pi^2 - m_\mu^2)^2 = f_\pi^4$ in the SUSY limit: the pion lifetime is set by the fourth power of the SUSY-breaking scale.

6 The supertrace sum rule

The supertrace of the squared mass matrix at the Seiberg vacuum receives contributions only from the soft SUSY-breaking masses (the W_{IJ}^2 piece cancels between scalar and fermion sectors):

$$\boxed{\text{STr}[M^2] = 18\,f_\pi^2 = 152{,}352\,\text{MeV}^2\,.} \tag{22}$$

The factor $18 = 2 \times 9$: two real components per complex meson, nine diagonal entries of the 3×3 meson matrix. This is an exact, parameter-free prediction.

In terms of hadronic spectroscopy:

$$\sum_{\text{scalar mesons}} m_S^2 - 2 \sum_{\text{mesinos}} m_F^2 = 18\, f_\pi^2 \,. \tag{23}$$

With 12 off-diagonal scalars at $2f_\pi^2$ each and 6 diagonal scalars at f_π^2 each: $\sum m_S^2 = 12 \times 2f_\pi^2 + 6 \times f_\pi^2 = 30\, f_\pi^2$. The mesino contribution (Section 7) is $\sum m_F^2 \approx 0.3\,\text{MeV}^2$, negligible. So $30f_\pi^2 - 12f_\pi^2 = 18f_\pi^2$ after subtracting the fermion mass contribution from the supertrace.

Testing this requires measuring the complete low-lying meson scalar spectrum and comparing with $18f_\pi^2$. The pion–muon relation already tests the off-diagonal piece; the diagonal piece at $f_\pi \approx 92\,\text{MeV}$ is the new content.

7 The mesino spectrum

7.1 Masses and hierarchy

The fermionic superpartners of the off-diagonal mesons form three Dirac pairs with masses set by the inverted seesaw:

Mesino pair	Spectator quark	m_{quark}	Mesino mass
$\psi(M_d^u, M_u^d)$	s	93.4 MeV	11 eV
$\psi(M_s^u, M_u^s)$	d	4.67 MeV	230 eV
$\psi(M_s^d, M_d^s)$	u	2.16 MeV	490 eV

The hierarchy is inverted relative to quark masses: the mesino pair involving the lightest quarks (u, d) has the *lowest* mass (spectator $= s$, the heaviest). The absolute scale is $m_\psi \sim \mu\Lambda^2/(m_i m_j) \sim 10\,\text{eV}$.

7.2 Long-range force from mesino exchange

The mesino Compton wavelengths define Yukawa force ranges:

Mesino	Mass	$\hbar c/m_\psi$	Range
$\psi(ud)$	11 eV	1.8×10^7 fm	1.8 cm
$\psi(us)$	230 eV	8.6×10^5 fm	0.86 mm
$\psi(ds)$	490 eV	4.0×10^5 fm	0.40 mm

These ranges (sub-mm to cm) are in the regime probed by torsion-balance fifth-force experiments. The Eöt-Wash experiment [13] bounds new Yukawa forces at $\lambda = 1\,\text{mm}$ to coupling $|\alpha| \lesssim 3 \times 10^{-4}$ relative to gravity.

The mesino–nucleon coupling is suppressed by the ISS seesaw ($\sim m_\psi/\Lambda$), but whether the resulting effective coupling exceeds the Eöt-Wash bound requires a detailed calculation of the mesino–nucleon vertex through the confined phase—a task analogous to computing the pion–nucleon form factor from QCD.

8 The Seiberg-dual sector: down-type quark masses

The Seiberg-magnetic dual of the SU(3) theory has meson VEVs $\langle M_j^j \rangle \propto 1/m_j$. The "magnetic charges" $\xi_j = 1/\sqrt{m_j}$ are the natural variables, and the Koide ratio $Q(\xi_d, \xi_s, \xi_b) = 0.6652$ deviates from 2/3 by only 0.22%. Setting this exactly to 2/3 and using one mass as input, the other two follow from the Koide quadratic:

Input	Predicted	PDG 2024	Deviation
(m_s, m_b)	$m_d = 4.60\,\text{MeV}$	$4.67 \pm 0.48\,\text{MeV}$	0.1σ
(m_d, m_b)	$m_s = 95.0\,\text{MeV}$	$93.4 \pm 0.8\,\text{MeV}$	1.9σ
(m_d, m_s)	$m_b = 4562\,\text{MeV}$	$4183 \pm 30\,\text{MeV}$	12.6σ

The m_b prediction fails because the Koide inversion amplifies the 0.22% Q-deviation into a large absolute error at extreme mass hierarchies. The m_d and m_s predictions are within 2σ, consistent with the magnetic interpretation: the down-type quarks organize as a near-Koide triple in the meson-condensate variables $1/\sqrt{m_j}$.

9 Summary of predictions and experimental programme

Quantity	sBootstrap	Experiment	Discrepancy	Section
f_π^{SUSY}	$91.19\,\text{MeV}$	$92.07 \pm 0.57\,\text{MeV}$	1.0%	2
m_π (off-diag.)	$\sqrt{2}\, f_\pi = 130\,\text{MeV}$	135–140 MeV	4–7%	3
$\lvert\langle\bar{q}q\rangle\rvert^{1/3}$	$276\,\text{MeV}$	$\sim 260\,\text{MeV}$	6%	3
GT discrepancy	0.35%	—	3.8× improved	4
$g_{\pi NN}$ (from GT)	13.12	13.17 ± 0.06	0.3%	4
m_μ	$104.9\,\text{MeV}$	$105.66\,\text{MeV}$	0.7%	5
m_σ^{sB}	$92\,\text{MeV}$	$f_0(500) \sim 450\,\text{MeV}$	open	3
$\text{STr}[M^2]$	$18\, f_\pi^2$	not yet measured	prediction	6
m_d (dual Koide)	$4.60\,\text{MeV}$	$4.67 \pm 0.48\,\text{MeV}$	0.1σ	8
m_s (dual Koide)	$95.0\,\text{MeV}$	$93.4 \pm 0.8\,\text{MeV}$	1.9σ	8

Table 1: Summary of nuclear and hadronic predictions.

The sBootstrap Lagrangian produces a coherent set of predictions at the nuclear scale:

1. The Goldberger–Treiman relation, with f_π^{SUSY} instead of f_π^{PDG}, is 3.8× more accurate. This is the cleanest nuclear test.

2. The pion–nucleon coupling $g_{\pi NN} = 13.12$ is a parameter-free output of the GT relation combined with the SUSY identification $f_\pi^2 = m_\pi^2 - m_\mu^2$. It matches experiment to 0.3%.

3. The chiral condensate $\lvert\langle\bar{q}q\rangle\rvert^{1/3} = 276\,\text{MeV}$ is a direct consequence of the GOR relation with SUSY inputs.

4. The muon mass is not free: $m_\mu = \sqrt{m_\pi^2 - f_\pi^2} = 104.9\,\text{MeV}$ (0.7% accuracy), connecting leptonic and hadronic scales through SUSY breaking.

5. The diagonal scalar mode at 92 MeV provides a medium-range nuclear attraction at 2.2 fm—a testable prediction for the NN potential.

6. The mesino spectrum (11–490 eV) predicts a new Yukawa force at sub-mm to cm range, accessible to fifth-force experiments.

7. The pion decay phase space is $\propto f_\pi^4$ in the SUSY limit: the pion lifetime is controlled by the fourth power of the SUSY-breaking scale.

The most urgent experimental tests are: (a) measuring or bounding a scalar-isoscalar state below 100 MeV; (b) improving fifth-force limits at 0.1–10 mm range; (c) lattice QCD computation of MSbar quark masses to 1% to test the dual Koide; (d) a precision comparison of $g_{\pi NN}$ and g_A to test whether the "correct" f_π for the GT relation is $\sqrt{m_\pi^2 - m_\mu^2}$.

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